

2.2 Programming fundamentals

How to run this unit, lesson by lesson

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Start here

8 lessons. A realistic estimate is **8 teaching hours**: most lessons are one hour, and a lesson with thirty or more activities takes two or three. The estimate counts transitions and assumes you teach from the front for the first fifteen minutes of each lesson.

Teach them in the order below. The order is the retrieval schedule as well as the teaching one: each starter asks about the lesson before it.

#	LESSON	HOURS	PLAN
1	Variable store (pf-101)	1	Lesson_Plans/PF_L01_LessonPlan.pdf
2	Type foundry (pf-102)	1	Lesson_Plans/PF_L02_LessonPlan.pdf
3	Filing room (pf-103)	1	Lesson_Plans/PF_L03_LessonPlan.pdf
4	Record vault (pf-104)	1	Lesson_Plans/PF_L04_LessonPlan.pdf
5	Array yard (pf-105)	1	Lesson_Plans/PF_L05_LessonPlan.pdf
6	Chance machine (pf-106)	1	Lesson_Plans/PF_L06_LessonPlan.pdf
7	Revision HQ 2.2 (pf-107)	1	Lesson_Plans/PF_L07_LessonPlan.pdf
8	End-of-topic test (pf-108-test)	1	Lesson_Plans/PF_L08_TEST_LessonPlan.pdf

Preparation and what you need

1. **Class logins, before the first lesson.** On the teacher dashboard (teacher.html), open **Class logins**, enter the class and the usernames one per line, and print the login cards. A pupil cannot sign in without one.
2. **Devices.** A laptop, a tablet or a phone will do. A headset is optional everywhere. The browser needs to be current; the experiences use WebGL.
3. **Print the worksheets** for each lesson. They are in this pack as .docx to edit and .pdf to print.
4. **Open each experience once yourself** on the teaching machine, so the images are cached before thirty pupils ask for them at the same moment.

Where progress is kept. In this build APP_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device a pupil used, not on a server. The teacher dashboard says so itself. Keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels with the pupil.

Pupils sign in at topics.html with the username and PIN on their card, choose the topic and then the lesson.

Using the three resources together

Slides for the front of the room; **the experience** for reading, deciding and being marked; **the worksheet** for writing and for the record. The rule that holds them together is write first, then answer at the badge.

Do not teach the station walls and then send pupils to read the same walls. Teach the first two or three from the slides and let them read the rest; the walls are there so a pupil can re-read what you said, not so you can say it twice.

Evidence, checking and common difficulties

Collect three things from every lesson: the starter (written), the key facts and station challenges on the worksheet (written), and the score screen breakdown (copied onto the worksheet by the pupil as soon as it appears). The score screen shows the marks per station, which is what tells you where to reteach; a single total does not.

The teacher dashboard shows progress by pupil and the class's weakest stations, averaged on first attempts. Use the weakest-station list to choose what to reteach rather than asking who found it hard.

Things to plan around in this unit

- only one activity asks the pupil to produce an answer rather than choose one — pf-101, pf-102, pf-103, pf-104, pf-105, pf-106, pf-107
- no optional challenge activity — pf-101, pf-102, pf-103, pf-104, pf-105, pf-106, pf-107

The lesson plans, in teaching order

Each lesson has its own plan as a separate PDF in Lesson_Plans/ of this pack, so you can print the one you are teaching. What follows is the summary of each; the plan carries the timings, the hinge question, the station-by-station delivery and the adaptations.

1. Variable store pf-101

How does a program store data, and how does it decide which line to run next?

OUTCOMES	know what variables and constants are, and how a value is assigned to each; understand how input and output move data in and out of a program; know what sequence, selection and iteration each do to the order a program runs in
STATIONS AND ACTIVITIES	6 stations, 14 activities, 21 marks (13 guided, 1 independent, 0 challenge)
RESOURCES	PF_L01_VariableStore_Lesson.pptx, PF_L01_VariableStore_Worksheet.docx
PLAN	Lesson_Plans/PF_L01_LessonPlan.pdf

2. Type foundry pf-102

Why does a program need to know what type of data it is working with?

OUTCOMES	know the data types integer, real, Boolean, character and string; understand why casting is needed and how a value is cast from one type to another; know how the arithmetic, comparison and Boolean operators are used
STATIONS AND ACTIVITIES	6 stations, 15 activities, 25 marks (14 guided, 1 independent, 0 challenge)
RESOURCES	PF_L02_TypeFoundry_Lesson.pptx, PF_L02_TypeFoundry_Worksheet.docx
PLAN	Lesson_Plans/PF_L02_LessonPlan.pdf

3. Filing room pf-103

How does a program keep hold of data after it has stopped running?

OUTCOMES	know why a program needs to store data in a file; understand how a file is opened, read from, written to and closed; know the difference between writing to a file and appending to it
STATIONS AND ACTIVITIES	6 stations, 14 activities, 18 marks (13 guided, 1 independent, 0 challenge)
RESOURCES	PF_L03_FilingRoom_Lesson.pptx, PF_L03_FilingRoom_Worksheet.docx
PLAN	Lesson_Plans/PF_L03_LessonPlan.pdf

4. Record vault pf-104

How does a program store a set of related data, and how does it search it?

OUTCOMES	know what is meant by a record, a field and a table of records; understand how SELECT, FROM and WHERE are used to search for data; know how AND, OR, NOT and LIKE narrow a search down
STATIONS AND ACTIVITIES	6 stations, 15 activities, 23 marks (14 guided, 1 independent, 0 challenge)
RESOURCES	PF_L04_RecordVault_Lesson.pptx, PF_L04_RecordVault_Worksheet.docx
PLAN	Lesson_Plans/PF_L04_LessonPlan.pdf

5. Array yard pf-105

How does one name hold many values, and how is code broken into reusable pieces?

OUTCOMES	know how one- and two-dimensional arrays are declared, read and changed; understand why arrays are zero-indexed and how a loop iterates over one; know the difference between a procedure and a function, and between local and global variables
STATIONS AND ACTIVITIES	6 stations, 14 activities, 19 marks (13 guided, 1 independent, 0 challenge)
RESOURCES	PF_L05_ArrayYard_Lesson.pptx, PF_L05_ArrayYard_Worksheet.docx
PLAN	Lesson_Plans/PF_L05_LessonPlan.pdf

6. Chance machine pf-106

How does a program produce a number that nobody can predict?

OUTCOMES	know what random numbers are used for in a program; understand how a random whole number is generated within a given range; know that a computer produces pseudo-random numbers from an algorithm and a seed
STATIONS AND ACTIVITIES	6 stations, 14 activities, 19 marks (13 guided, 1 independent, 0 challenge)
RESOURCES	PF_L06_ChanceMachine_Lesson.pptx, PF_L06_ChanceMachine_Worksheet.docx
PLAN	Lesson_Plans/PF_L06_LessonPlan.pdf

7. Revision HQ 2.2 pf-107

Which parts of 2.2 am I secure on, and which do I need to revise before the test?

OUTCOMES	know the whole of topic 2.2 well enough to answer without notes; understand the mistakes that most often lose marks in programming questions; know which lessons are secure and which still need revision before the test
STATIONS AND ACTIVITIES	6 stations, 15 activities, 20 marks (14 guided, 1 independent, 0 challenge)
RESOURCES	PF_L07_RevisionHQ22_Lesson.pptx, PF_L07_RevisionHQ_Worksheet.docx
PLAN	Lesson_Plans/PF_L07_LessonPlan.pdf

8. End-of-topic test pf-108-test

Your 2.2 test is done on paper. Revise with Revision HQ 2.2 first, then use this reflection sheet once it's marked.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a test and reflection sheet. It assesses the outcomes taught across the unit.
STATIONS AND ACTIVITIES	0 stations, 0 activities, 0 marks (0 guided, 0 independent, 0 challenge)
RESOURCES	PF_L08_EndOfTopicTest_Reflection_Worksheet.docx
PLAN	Lesson_Plans/PF_L08_TEST_LessonPlan.pdf

Teacher answers

Model answers and marking guidance are not in this guide and not on the lesson plans. A plan is printed and left on a desk; an answer sheet is not. Where verified answers exist for this unit they are in the Teacher_Answers/ folder of this pack. Where they do not, the pack says so rather than shipping an empty file.

For the programming lessons there is no single model answer to print: a program is marked by running it against its test cases, and many different correct programs pass. The tests themselves are the marking guidance, and they are in the teacher answers.

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